

Micro-AGI: Emergent Intelligence from Networks of Small Language Models

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February 2026

Abstract

We present **Micro-AGI**, an architecture where multiple specialized LoRA adapters share a single frozen language model and communicate through tensor-level signals rather than natural language. A router network activates different cognitive specialists (reasoner, critic, imagineer, planner, synthesizer) based on input characteristics, and inter-agent communication occurs through the shared KV cache. Across five iterations (v0.1–v0.5), we report both successes and failures: routing-augmented responses scored up to **+1.6 points** higher than baseline when evaluated by Claude Opus, but the gains vanished when using a smaller model as both respondent and judge. We provide a detailed analysis of *why* the architecture works for strong models and fails for weaker ones, and identify the fundamental bottleneck: the LM head compresses rich internal representations into surface-level token probabilities, discarding the signal that makes multi-agent coordination useful. This negative result points toward vector-based communication as the necessary next step.

1 Introduction

The prevailing approach to multi-agent AI involves running separate large language model instances that communicate through text. This is computationally expensive (each agent requires a full model in memory) and informationally lossy (rich internal representations are compressed into tokens for communication).

We explore an alternative: what if multiple “agents” could share a single model, switching between specialized behaviors through LoRA adapters, and communicating through internal representations rather than text?

The Micro-AGI architecture implements this

idea using:

- One frozen Qwen2.5-1.5B model (~1.3 GB in 4-bit)
- Five LoRA adapters (~36 MB each) for specialized cognitive functions
- Hot-swapping between adapters in ~2.7ms
- Total memory: ~1.5 GB—less than a single 7B model

1.1 Contributions

1. A complete **multi-agent architecture on a single GPU** with 5 specialized agents sharing one base model.
2. **Five iterations** of the architecture (v0.1–v0.5) with progressive improvements and honest reporting of failures.
3. A **critical analysis** of why top-token signals are architecturally unsuitable for inter-agent communication.
4. Evidence that the **respondent model’s capability** determines whether routing signals add value—not the signals themselves.

2 Architecture

2.1 Base Model and Adapters

The frozen base model (Qwen2.5-1.5B-Instruct, 4-bit NF4 quantization) serves as the shared “brain.” Five LoRA adapters are trained on 150 domain-specific examples each:

Table 1: Specialized LoRA adapters

Agent	Role	r	Val Loss
Reasoner	Causal analysis	16	1.46
Critic	Risk/flip detection	16	1.56
Imagineer	Creative solutions	8	1.76
Planner	Action sequencing	8	1.68
Synthesizer	Integration	8	1.75

2.2 Processing Pipeline

For each input query:

- Initial activation:** The input is processed through each adapter sequentially.
- Signal extraction:** After each adapter pass, we extract the top- k predicted tokens and their probabilities, plus the confidence delta between iterations.
- Routing:** Based on confidence evolution, the system selects which agents to re-activate for additional iterations.
- Signal compilation:** The accumulated signals (agent order, confidence evolution, top tokens) are formatted as context for a responding model.

Hot-swapping between LoRA adapters takes ~ 2.7 ms, enabling rapid cycling through all five agents in under 20ms.

2.3 Confidence-Based Routing

The router monitors prediction confidence across iterations:

$$c_t = \max_i p_t(w_i) - \max_i p_{t-1}(w_i) \quad (1)$$

where $p_t(w_i)$ is the probability of the i -th token at iteration t . When confidence stabilizes ($|c_t| < \epsilon$), the system stops iterating. When confidence drops, it activates additional agents.

3 Experimental Evolution

3.1 Version History

Table 2: Architecture evolution and benchmark results

Ver.	Base	Aug.	Δ	Judge
v0.2	6.8	8.0	+1.2	Opus
v0.3	6.6	8.2	+1.6	Opus
v0.4	7.0	8.3	+1.4	Opus
v0.5	7.0	6.9	-0.1	Opus [†]

[†] v0.5: Opus as judge, Qwen-7B as respondent

3.2 Key Improvements Across Versions

v0.1: Single LoRA (reasoner), no routing. Basic proof of concept.

v0.2: Four adapters (reasoner, critic, imagineer, planner), sequential activation. First positive results (+1.2).

v0.3: Added soft prompts (Thought Engine integration) and iterative routing. Best delta (+1.6).

v0.4: Confidence-based routing with early stopping. More efficient (fewer iterations) with comparable quality (+1.4).

v0.5: Five adapters trained on 750 examples (5×150), expanded LoRA rank. Despite better training, *delta collapsed to zero* when switching from Claude to Qwen as respondent.

4 The Critical Failure

4.1 What Went Wrong in v0.5

The v0.5 benchmark used Qwen-7B (via Ollama) as the respondent instead of Claude Opus. The routing signals were identical—same agents, same confidence patterns, same top tokens. But the quality delta vanished.

4.2 Root Cause Analysis

We identified two fundamental issues:

1. The LM Head Bottleneck. The “signals” extracted from each agent pass are the top predicted tokens—the output of the LM head layer. But the LM head is optimized for next-token prediction in natural language. Regardless of what the model “thinks” internally, the most probable tokens are always function words: ``La'', ``Le'', ``Les'', ``-', ``#''.

The rich internal representations (hidden states of dimension 1,536) are compressed into a probability distribution over 151,936 vocabulary tokens. This compression discards exactly the information that makes agent specialization useful.

2. Respondent Capability Determines Value. Claude Opus is capable of extracting meta-information from noisy signals. It interprets “critic activated first” as “this question has ethical implications” without being told. Qwen-7B takes the signals literally, attempting to reason about “La(3%), Le(3%)” as meaningful concepts. The signals are useful only if the consumer is smart enough to interpret them.

4.3 What the Signals Actually Capture

Despite the token-level noise, the routing-level signals contain real information:

- **Agent activation order:** Critic-first for ethical questions, planner-first for strategic ones.
- **Confidence evolution:** High uncertainty (many iterations) for complex/ambiguous questions, quick convergence for straightforward ones.
- **Iteration count:** Strong correlation between required iterations and question complexity.

5 Lessons Learned

5.1 Negative Results as Progress

The v0.5 failure is arguably more valuable than the v0.3 success. It demonstrated conclusively that:

1. More training data (30→150 examples) does not fix an architectural problem.
2. Larger LoRA rank (8→16) does not produce semantically meaningful top tokens.
3. The problem is not the agents—it is the communication channel (token probabilities).

5.2 The Communication Medium Matters

Three levels of inter-agent communication were identified during this research:

Table 3: Communication density hierarchy

Medium	Density	Status
Natural language	1×	Baseline
.ava notation	3–6×	Tested
Hidden states/vectors	Max	Needed

The .ava notation (a compressed symbolic format developed in parallel work) achieves 3–6× compression over natural language but still operates in the token space. The true next step is communication through hidden states or learned vectors.

6 Related Work

Mixture of Experts (Shazeer et al., 2017): MoE routes inputs to specialized sub-networks. Micro-AGI is similar in spirit but uses full LoRA adapters instead of expert layers, and operates at the agent level rather than the layer level.

Multi-Agent Debate (Du et al., 2023): Agents debate through text to improve reasoning. Our work shows that text-based communication is the bottleneck, not the solution.

KV Cache Sharing: To our knowledge, no published work shares KV cache between LoRA-adapted agents on a single model as a communication mechanism. This appears to be a novel configuration.

7 Conclusion

Micro-AGI demonstrates that a multi-agent cognitive architecture can run on a single consumer GPU with remarkable efficiency (5 agents, 1.5 GB, 2.7ms swap). The routing signals capture genuine information about question complexity and type. However, the LM head bottleneck prevents this information from being communicated effectively through token probabilities.

The path forward is not better LoRA training or more data—it is fundamentally changing the communication medium from tokens to vectors. The Thought Engine (companion paper) provides initial evidence that learned vectors can steer cognition more effectively than text. The next step is a complete vector-based communication protocol between agents.

Reproducibility

All code, datasets, and trained adapters are available at: <https://gitlab.avalon-network.com/acros/ava> (directory: experiments/neural-network/).

References

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